stmtext class description

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`stmtext` is a wrapper documentclass around a selection of existing class layouts for LaTeX. Additionally, it allows quick access to the modules of `stmlatex`.

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1. About

`stmtext` is supposed to be an easy-to-use interface to several other document classes with the additional use of the modules of `stmlatex`. The class addresses users who want to use certain features without initially worrying about their implementation. However, there will most certainly come a point in the learning curve of a LaTeX user where this documentclass becomes insufficient. That's what's wanted. Maybe, until this point, the class has helped to bring some possibilities of LaTeX closer.

The goal is to make LaTeX functionalities available to the user. It is not intended to make any changes to the document layout, despite loading the underlying document classes. The rest of the layout can be freely changed by the user.

The complete class and module system can be found at
https://github.com/raedma/stmlatex

2. Options

This section describes the class options. All options are key-value-options, meaning they are set using the definition `KEY=VALUE`. It must be noted that these options can not represent the full features of the module packages. In case more advanced options are required make sure to have a look at the individual package descriptions and consider loading them individually. Maybe you find what you need.

2.1. Overview

This overview contains the most important informations about package options. If there are additional comments about individual options an additional section is created for their description.

Boolean options These options can only have the values `true` or `false`.

Table 1: Boolean class options

Key	Default value	Description
`attachfile`	`false`	Load the `attachfile` package at the correct position relative to `stmhyperref`
`bibliography`	`false`	Load the `bibliography` package
`calc`	`false`	Load the `calc` package at the correct position relative to `stmtikz`
`cleverref`	`false`	Load the `cleveref` package at the correct position relative to `stmhyperref`
`color`	`true`	Load the `stmcolors` package
`conditionallox`	`true`	Load the `stmconditionallox` package

continued ...

. . . continued

Key	Default value	Description
conditionalloxnewpage	true	Set the value of the newpage option in the stmconditionallox package
date	false	Load the stmdate package
glossaries	false	Load the stmglossaries package
index	false	Load the stmindex package
hyperref	true	Load the stmhyperref package
listings	false	Load the stmlistings package
math	false	Load the stmmath package
plots	false	Load the stmplots package
tabular	false	Load the stmtabular package
tikz	false	Load the stmtikz package
units	false	Load the stmunits package

String options These options have strings as values. Possible arguments are listed. The default is underlined.

Table 2: String class options

Key	Values	Description
language	german|english	Document language, Load the stmlanguage package with this option
layout	standard|dlr|koma	Document layout
type	article|report|book	Base document type

The following options are only available for layout=dlr which uses the RM-LaTeX based document classes.

Table 4: Additional string class options for layout=dlr

Key	Values	Description
institute	FA-STM	See the RM-LaTeX documentation for alternatives
site	BS	See the RM-LaTeX documentation for alternatives

2.2. Option attachfile

The package attachfile is one of the packages that has to be loaded at a specific relative position to hyperref. Thus, this option is only relevant in case the Option hyperref from section 2.7 is used.

2.3. Option `cleveref`

The package `cleveref` is one of the packages that has to be loaded at a specific relative position to `hyperref`. Thus, this option is only relevant in case the Option `hyperref` from section 2.7 is used.

2.4. Option `conditionallox`

In case the option `conditionallox` is used, the document has to be compiled at least three times.

2.5. Option `glossaries`

Setting `glossaries=true` loads the `stmglossaries`. Thus, the user is able to use the glossary entries from the package as described in the package documentation.

In case the `arara` package is used for the build automation of the LaTeX document, add

```
arara: makeglossaries
```

in the build automation part of your preamble. That way, the glossaries are automatically compiled during your document creation.

For the glossaries to show up in your document, simply place the commands to print the glossary entries of your choice at the location in your document where you want them to show up, e.g.

```
\printstmacronyms

\printstmscalarglossary
\printstmvectorglossary
\printstmmatrixglossary
\printstmstateglossary
\printstmindexglossary
\printstmexponentglossary
\printstmoperatorglossary
```

or

```
\printstmacronyms
\printallstmsymbols
```

In case no entries are defined, the lists are simply omitted in your document.

2.6. Option `index`

2.6.1. Base package issues

There are issues with the `nonewpage` option of the base package. Thus, currently, there is no chance to disable that a index is created on a new page.

2.6.2. Individual indices

In case it is desired to use other than the default index settings, e.g. create a new index next or replacing the standard, it is recommended to load `stmtext` with the option `index=false` and loading the index individually. Otherwise, there might be conflicts in case `tikz=true` is used in combination.

```
\documentclass[%
  ...
  index=false,%
  ...
]{stmtext}

...

% replace standard index
\usepackage[
  standard=false
]{stmindex}

...

% Must come after \makeindex, e.g. from stmindex
\usepackage{stmtikzexternalization}
```

2.7. Option `hyperref`

https://tex.stackexchange.com/a/26592

2.8. Option `layout`

This class is supposed to be an interface between different class bundles. Currently, this class offers an interface to the standard LaTeX class bundle (`article`, `report`, `book`), the KOMA-script bundle (`scrartcl`, `scrreprt`, `scrbook`) and the RM-LaTeX bundle.

It is not the goal to implement all options of all classes rather than to allow an easy switch for standard documents. In case you miss an option for a specific bundle, feel free to implement it.

2.9. Other options

All other options not explicitly list in section 2.1 are passed to the underlying document class. Thus, it has to be made sure that the underlying document class understands these options. At this point, switching the argument in the Option `layout` may lead to problems in case the options is specific to a document class.

3. Usage

3.1. `type`-independent document structure

coseoul package

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A. The code

A.1. stmtext.cls

```
%%%
```

```
40
41 % --------------------------
42 % Options base
43 % --------------------------
44
45 \SetupKeyvalOptions{%
46   family=stmtextbase,%
47   prefix=stmtextbase@,%
48   setkeys=\kvsetkeys,%
49 }
50
51 % String options
52 \DeclareStringOption[report]{type}
53 \DeclareStringOption[koma]{layout}
54
55 % Process options
56 \ProcessKeyvalOptions{stmtextbase}
57
58 % --------------------------
59 % Options
60 % --------------------------
61
62 \SetupKeyvalOptions{%
63   family=stmtext,%
64   prefix=stmtext@,%
65   setkeys=\kvsetkeys,%
66 }
67
68 % Bool option
69 \DeclareBoolOption[false]{bibliography}
70 \DeclareBoolOption[true] {color}
71 \DeclareBoolOption[true] {conditionallox}
72 \DeclareBoolOption[true] {conditionalloxnewpage}
73 \DeclareBoolOption[false]{date}
74 \DeclareBoolOption[false]{glossaries}
75 \DeclareBoolOption[false]{index}
76 \DeclareBoolOption[true] {indexstandard}
77 \DeclareBoolOption[true] {hyperref}
78 \DeclareBoolOption[false]{listings}
79 \DeclareBoolOption[false]{math}
80 \DeclareBoolOption[false]{plots}
81 \DeclareBoolOption[false]{tabular}
82 \DeclareBoolOption[false]{tikz}
83 \DeclareBoolOption[true] {tikzexternalization}
```

```
\DeclareBoolOption[false]{units}

%
\DeclareStringOption[english]{language}

% DLR options
\DeclareStringOption[SY-STM]{institute}
\DeclareStringOption[BS]{site}

% Options that are dependent of the position of hyperref
\DeclareBoolOption[false]{attachfile}
\DeclareBoolOption[false]{cleveref}

% Options that are dependent of the position of tikz
\DeclareBoolOption[false]{calc}

% any other options are passed to the base class
\DeclareDefaultOption{%
  \ifthenelse{\equal{\stmtextbase@type}{article}}{%
    \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
      \PassOptionsToClass{\CurrentOption}{article}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
      \PassOptionsToClass{\CurrentOption}{dlrartcl}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
      \PassOptionsToClass{\CurrentOption}{scrartcl}%
    }{}%
  }{}%
}
\DeclareDefaultOption{%
  \ifthenelse{\equal{\stmtextbase@type}{report}}{%
    \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
      \PassOptionsToClass{\CurrentOption}{report}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
      \PassOptionsToClass{\CurrentOption}{dlrreprt}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
      \PassOptionsToClass{\CurrentOption}{scrreprt}%
    }{}%
  }{}%
}
\DeclareDefaultOption{%
```

```
  \ifthenelse{\equal{\stmtextbase@type}{book}}{%
    \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
      \PassOptionsToClass{\CurrentOption}{book}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
      \PassOptionsToClass{\CurrentOption}{dlrbook}%
    }{}%
    \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
      \PassOptionsToClass{\CurrentOption}{scrbook}%
    }{}%
  }{}%
}

% Process options
\ProcessKeyvalOptions{stmtext}

% ----------------------------
% Base class
% ----------------------------

\ifthenelse{\equal{\stmtextbase@type}{article}}{%
  \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
    \LoadClass{article}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
    \LoadClass{dlrartcl}
    \RequirePackage{auxhook}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
    \LoadClass{scrartcl}
    \RequirePackage{auxhook}
  }{}%
}{}%
\ifthenelse{\equal{\stmtextbase@type}{report}}{%
  \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
    \LoadClass{report}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
    \LoadClass{dlrreprt}
    \RequirePackage{auxhook}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
    \LoadClass{scrreprt}
    \RequirePackage{auxhook}
```

```
  }{}%
}{}%
\ifthenelse{\equal{\stmtextbase@type}{book}}{%
  \ifthenelse{\equal{\stmtextbase@layout}{standard}}{%
    \LoadClass{book}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%
    \LoadClass{dlrbook}
    \RequirePackage{auxhook}
  }{}%
  \ifthenelse{\equal{\stmtextbase@layout}{koma}}{%
    \LoadClass{scrbook}
    \RequirePackage{auxhook}
  }{}%
}{}%

% ---------------------------
% Commands for communality between document classes
% ---------------------------

% Commands for the standard document classes
\ifthenelse{\equal{\stmtextbase@layout}{standard}}{%

  \@ifundefined{subtitle}{    \def\subtitle#1{\gdef\@subtitle{#1} }%
                              \def\@subtitle{}}{}%

}{}%

% Commands for DLR document classes
\ifthenelse{\equal{\stmtextbase@layout}{dlr}}{%

  \dlrinstitute{\stmtext@institute}%
  \dlrsite     {\stmtext@site}%

}{}%

% Commands from RM-LaTeX that other classes need to understand
\ifthenelse{%
  \equal{\stmtextbase@layout}{standard} \OR%
  \equal{\stmtextbase@layout}{koma}%
}{%

    % These commands are created to allow for an easy switch between
    % dlrreprt and bootstrap_dlrreprt.
```

```
216
217     \@ifundefined{subject}{     \def\subject#1{\gdef\@subject{#1} }%
218                                 \def\@subject{}}{}%
219     \@ifundefined{keywords}{    \def\keywords#1{\gdef\@keywords{#1} }%
220                                 \def\@keywords{}}{}
221     \@ifundefined{restrictions}{\def\restrictions#1{\gdef\@restrictions
        {#1} }%
222                                 \def\@restrictions{}}{}
223     \@ifundefined{author}{      \def\author#1{\gdef\@author{#1} }%
224                                 \def\@author{}}{}
225     \@ifundefined{coauthor}{    \def\coauthor#1{\gdef\@coauthor{#1} }%
226                                 \def\@coauthor{}}{}
227     \@ifundefined{savedby}{     \def\savedby#1{\gdef\@savedby{#1} }%
228                                 \def\@savedby{}}{}
229     \@ifundefined{approvedby}{  \def\approvedby#1{\gdef\@approvedby{#1}
        }%
230                                 \def\@approvedby{}}{}
231     \@ifundefined{releasedby}{  \def\releasedby#1{\gdef\@releasedby{#1}
        }%
232                                 \def\@releasedby{}}{}
233     \@ifundefined{contactname}{ \def\contactname#1{\gdef\@contactname
        {#1} }%
234                                 \def\@contactname{}}{}
235     \@ifundefined{telnumber}{   \def\telnumber#1{\gdef\@telnumber{#1} }%
236                                 \def\@telnumber{}}{}
237     \@ifundefined{faxnumber}{   \def\faxnumber#1{\gdef\@faxnumber{#1} }%
238                                 \def\@faxnumber{}}{}
239     \@ifundefined{email}{       \def\email#1{\gdef\@email{#1} }%
240                                 \def\@email{}}{}
241     \@ifundefined{reportno}{    \def\reportno#1{\gdef\@reportno{#1} }%
242                                 \def\@reportno{}}{}
243
244     % From dlrinstitutes.sty
245     \@ifundefined{institute}{   \def\institute#1{\gdef\@institute{#1} }%
246                                 \def\@institute{}}{}
247     \@ifundefined{site}{        \def\site#1{\gdef\@site{#1} }%
248                                 \def\@site{}}{}
249
250     % The command for the second titlepage
251     \newcommand{\secondpage}{}%
252     \newcommand{\printsecondpage}{}%
253 }{
254
255   \RequirePackage{dlrsecondpage}
```

```
256
257 }
258
259 % --------------------------
260 % Base packages
261 % --------------------------
262
263 \PassOptionsToPackage{%
264   language=\stmtext@language,%
265 }{stmlanguage}%
266 \RequirePackage{stmlanguage}
267
268 % Must come after language
269 \ifstmtext@bibliography
270   \RequirePackage{stmbibliography}
271 \fi
272
273 \ifstmtext@color
274   \RequirePackage{stmcolors}
275 \fi
276
277 \ifstmtext@conditionallox
278   \RequirePackage[%
279     newpage={\ifstmtext@conditionalloxnewpage true\else false\fi},%
280   ]{stmconditionallox}
281 \fi
282
283 \ifstmtext@date
284   \RequirePackage{stmdate}
285 \fi
286
287 % This must be loaded ahead of tikz externalization
288 \ifstmtext@index
289   \PassOptionsToPackage{%
290     standard={\ifstmtext@indexstandard true\else false\fi},%
291   }{stmindex}%
292   \RequirePackage{stmindex}
293 \fi
294
295 \ifstmtext@listings
296   \RequirePackage{stmlistings}
297 \fi
298
299 \ifstmtext@math
```

```
  \RequirePackage{stmmath}
\fi

\ifstmtext@plots
  \RequirePackage{stmplots}
\fi

\ifstmtext@tabular
  \RequirePackage{stmtabular}
\fi

% Must be loaded before tikz
\ifstmtext@calc
  \RequirePackage{calc}
\fi

\ifstmtext@tikz
  \PassOptionsToPackage{%
    externalization={\ifstmtext@tikzexternalization true\else false\fi},
    %
  }{stmtikz}%
  \RequirePackage{stmtikz}
\fi

\ifstmtext@units
  \RequirePackage{stmunits}
\fi

% ---------------------------
% Packages with dependencies
% ---------------------------

% ---------------------------
% Coseoul to get a level-independent document structure
% ---------------------------

\RequirePackage{coseoul}

\newcommand{\basetoclevelnr}[1]{#1}
\ifthenelse{\equal{\stmtextbase@type}{article}}{%
  \renewcommand{\basetoclevelnr}{5}%
}{}%
\ifthenelse{\equal{\stmtextbase@type}{report}}{%
  \renewcommand{\basetoclevelnr}{6}%
```

```
}{}%
\ifthenelse{\equal{\stmtextbase@type}{book}}{%
  \renewcommand{\basetoclevelnr}{7}%
}{}%

\setcounter{currentlevel}{\basetoclevelnr}

% ---------------------------
% Commands dependent of the document base type
% ---------------------------

\ifthenelse{\equal{\stmtextbase@type}{article}}{%

}{}%
\ifthenelse{\equal{\stmtextbase@type}{report}}{%

}{}%
\ifthenelse{\equal{\stmtextbase@type}{book}}{%

  % http://mirrors.ctan.org/macros/latex/contrib/biblatex/doc/biblatex.
    pdf#subsubsection.3.1.2
  \ExecuteBibliographyOptions{%
    refsection=part,% Individual bibliograpy for each part
  }

}{}%

% ---------------------------
% Put stuff in the TOC
% ---------------------------

\ifthenelse{%
  \equal{\stmtextbase@layout}{dlr} \OR%
  \equal{\stmtextbase@layout}{koma}%
}{%
  \KOMAoptions{%
    bibliography=totoc,%
    %figures=plain,%
    %fleqn,% not supported in KOMA options
    listof=totoc,%
  }%
}{}%

\ifthenelse{%
```

```
  \equal{\stmtextbase@layout}{standard}%
}{%

  %\@ifpackageloaded{biblatex}{
    %\PassOptionsToPackage{toc=bib}{biblatex}
  %}{%
    \PassOptionsToPackage{%
      nottoc,%
      notbib,%
    }{tocbibind}
    \RequirePackage{tocbibind}
  %}
}{}%

% ----------------------------
% Hyperref - make sure it is loaded as the last package
% ----------------------------

\ifstmtext@hyperref
  %\makeatletter

  \@ifpackageloaded{etoolbox}{}{%
    \RequirePackage{etoolbox}
  }

  %\AtBeginDocument{
  \AtEndPreamble{

    %
    \RequirePackage{stmhyperref}

    % After hyperref for links to work
    % dependency on babel language
    \ifstmtext@glossaries
      \RequirePackage{stmglossaries}
    \fi


    \ifstmtext@attachfile
      \RequirePackage{attachfile}
    \fi

    \ifstmtext@cleveref
      \RequirePackage{cleveref}
```

```
    \fi
  }
  %\makeatother
\else
  \ifstmtext@attachfile
    \RequirePackage{attachfile}
  \fi

  \ifstmtext@cleveref
    \RequirePackage{cleveref}
  \fi
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                              %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```